

SPATIAL DISTRIBUTION AND DEVELOPMENT OF ROOT SPROUTS IN *FAGUS GRANDIFOLIA* (FAGACEAE)¹

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ABSTRACT

Root sprouts around 31 *Fagus grandifolia* parent trees, some declining due to beech bark disease, were studied to describe the pattern of sprout distribution, the ecological importance of this pattern, and the relationship between sprout patterns and parent vigor. Microtome sections of roots were studied to determine the histological origin of sprouts. Sprout distribution was circular and usually centered on the parent tree. Most sprouts were within 8 m of the parent and remained attached to the parent root system even after the sprouts reached 10 yrs of age. Root sprouting in *F. grandifolia* may be effective in replacing dead parent trees, but the potential for clones to spread is limited. Spatial distribution of sprouts was mostly unaffected by tree vigor. Root sprouts originated from callus tissues associated with wounds on superficial woody roots.

VEGETATIVE SPREAD by layering, rhizome production, stump sprouting, or root sprouting, occurs in many woody, north-temperate plants (Fowells, 1965; Barnes, 1966; Vasek, 1980; Légère and Payette, 1981; Huenneke, 1985). The result of such reproduction is a group of connected or separate individual ramets, the sum of which constitutes a single genet (Kays and Harper, 1974). The degree of ramet connection and the spatial-temporal patterns of ramet distribution are thought to be important to genet fitness (Cook, 1983; Bell, 1984).

A common pattern for ramet distribution in early-successional or shade-intolerant woody species is a circle with the tallest and oldest plants at the center and the younger plants on the outer margin (Barnes, 1966). The circular arrangement of ramets expands until certain competing plants or environmental barriers are encountered. Eventually, older ramets in the center may senesce, and if such senescence is repeated over time, a distinct pattern of centrifugal or wave-like genet movement may result (Watt, 1947; Vasek, 1980).

Patterns of clone development in woody shade-tolerant species are less well known, partly because few of them form clones. *Fagus*

grandifolia Ehrh. (American beech) is a shade-tolerant tree common to eastern North America and is known to reproduce vegetatively by root sprouts (Rushmore, 1961). Although circular spatial patterns of beech sprouts in closed-canopied forests have been observed, published information is restricted to just one report (Ward, 1961) on beech clones growing in eastern Wisconsin. Ward found that: 1) most sprouts originate from injured woody roots near the soil surface and within 7.3 m of the parent; 2) sprouts are more common on slopes; and 3) the number of sprouts is correlated with parent size (although many trees without any sprouts were noted). No detailed studies on temporal distributions of beech ramets have been reported; thus, the potential for genet movement in beech is poorly understood.

The morphology and histological origins of newly-initiated beech sprouts are also poorly understood. Studies of other species indicate the occurrence of two general patterns: 1) development of sprouts from non-apical meristems in a single growing season (e.g., as in *Populus tremuloides* Michx. [Schier, 1981]); and 2) formation of adventitious buds that may remain dormant for many years before expanding to form shoots (e.g., as in *Liquidambar styraciflua* L. [Kormanik and Brown, 1967; Hook, Kormanik and Brown, 1970; Francis, 1982]).

The recent development of the beech bark disease, a decline disease (sensu Manion, 1981), has accentuated the need to understand the pattern and ecological significance of beech regeneration. The disease is initiated by a scale insect, *Cryptococcus fagisuga* Lind., which al-

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lows facultative parasitic fungi (*Nectria cocinea* var. *faginata* Loh., Wats., and Ayers and *N. galligena* Bres.) to develop in the cortex and cambium of the tree. In infected stands, the disease may result in heavy mortality, especially among larger trees (Shigo, 1972; Houston, 1975; Miller-Weeks, 1983; Twery and Patterson, 1984). Of specific interest is the potential for future severe outbreaks of the disease if parent mortality stimulates prolific root sprouting of susceptible genotypes (Houston, 1975).

These concerns have prompted our study to address several questions including: 1) what is the spatial distribution and degree of connection of beech root sprouts in closed canopy forests; 2) what are the ecological implications of this distribution; 3) what changes, if any, occur in the spatial distribution of sprouts when parent vigor declines due to beech bark disease; and 4) from what tissues do sprouts develop?

This study tests specific hypotheses of spatial distribution, developed from preliminary field observations and observations by Ward (1961), including: 1) sprouts form a circle around the parent tree with the youngest sprouts at the periphery; 2) sprout populations are centered on the downslope side of the parent, with the distance between the parent and the sprout centroid positively correlated with percent slope; and 3) sprout density is closely correlated with the density of woody roots located above the mineral soil surface.

METHODS—Sample selection and measurement of variables—Sampling was conducted in a portion of the glaciated Allegheny Plateau region of central New York including northern Cortland, northern Chenango, and southern Madison Counties. The region has cool summers and cold winters with an average frost-free period of approx. 120 days (N.O.A.A., 1978). An average annual precipitation of 950 mm is well distributed throughout the year. Mean annual snowfall ranges from 175 to 230 cm. Sampling was restricted to Dystrochrept and Fragiochrept soils developed in lodgement till composed of mixed shale, siltstone, and sandstone fragments (Hanna, 1981; Crandall, 1985).

For spatial analyses, 24 × 24 m plots were located in 13 closed canopy northern hardwood and hemlock-northern hardwood stands having no evidence of logging for many years (Jones, 1986). The predominant species in a composite of the sampled stand, ranked according to basal area, were *Acer saccharum* Marsh., *A. rubrum* L., *Prunus serotina* Ehrh., *Tsuga canadensis* (L.) Carr., *Fagus grandifolia*,

and *Fraxinus americana* L. Single parent beech trees with diameters at 1.37 m (DBH) of more than 15 cm were chosen as plot centers. To avoid confusing parental affiliation of sprouts, each parent tree was at least 20 m from any other beech tree having a DBH over 10 cm. Based on these criteria, 31 plots were selected and sampled in late summer through autumn 1983 (13 plots), summer 1984 (8 plots), and summer 1985 (10 plots).

In each plot, two axes (X = west to east, Y = south to north) were laid out and used to map the location of sprouts, seedlings, and trees. For all trees (woody stems more than 3 cm in DBH) in the plot, DBH and map coordinates were measured. Plot aspect was determined with a compass and plot slope was measured using a clinometer. A soil pit, dug at the edge of the plot, was used to determine the soil series and to describe the soil profile.

For parent trees, the following were recorded: DBH; crown health index (0 = healthy; 1 = chlorotic, or with a few small dead branches; 2 = some major branches recently dead, but more than 50% of branches still alive; 3 = more than 50% of branches recently dead); percent cover of the bole (up to the first major branch) by recently formed *Nectria* spp. perithecia and by *Cryptococcus fagisuga* infestation; and average annual radial growth for the past 20 yrs determined from five increment cores at DBH.

For all beech stems other than the parent tree, the following data were collected: origin (sprout, seedling, or unknown), map coordinates, and total ht in cm. Origin was determined by excavating the base of each stem. A large 1984 seed crop resulted in numerous 1-yr old seedlings in 1985; consequently, 1-yr old seedlings were counted but not mapped in one 1985 plot. In many cases, two or more sprouts arose from locations within 5 cm of each other on a root. To expedite field work, each parent root location with one or more sprouts attached was treated as an individual sprout clump and map coordinates were recorded for the tallest sprout per clump (TSPC) only. In all 31 plots, each TSPC was aged by counting bud scale scars for sprouts less than 9 yrs old, or by xylem ring counts for sprouts 9 yrs old or older. Additional measurements, made only in the 21 plots sampled in 1983–84, included: 1) annual shoot extension of each TSPC, estimated by measuring the length of annual shoot growth of the tallest branch for 1980–1983; 2) depth of origin within soil horizons of each TSPC; 3) position of TSPC on the parent root, classified as top, side, or bottom; 4) root diam on either side of each sprout clump; 5) presence or absence of an apparent

connection to the parent; and 6) number of living sprouts per clump.

The density and distribution of woody roots at or above the mineral soil surface (hereafter called superficial roots following Sutton and Tinus [1983]), were estimated by: 1) systematically locating fifty-two 0.25 m² quadrats around the 21 parent trees sampled in 1983 and 1984; and 2) measuring within each quadrat the lengths and mean diam of all superficial roots 0.5 cm or more in diam.

Analyses—Visual inspection and the product moment correlation of X and Y coordinate axes were used to determine if sprouts were distributed in a circle or in an ellipse. To see if new sprouts were clustered on the perimeter of the clone, analysis of variance was used to test for differences among age classes in mean sprout-to-parent distance.

The hypothesis that sprout distributions were offset from the parent was tested using Hotelling's multivariate t statistic (Morrison, 1976) which tests for a significant departure of the mean X and Y coordinates ($X = \bar{x}$, $Y = \bar{y}$) from the origin ($X = 0$, $Y = 0$). The hypothesis that sprout populations are centered on the downslope side of the parent was tested by: 1) estimating an expected angle for uncorrelated vectors by randomly choosing 1,000 pairs of azimuths and by computing the mean angle (0–180°) formed by the random pairs; 2) computing the mean angle formed by the aspect (azimuth) and the vector ($\bar{x}$, $\bar{y}$) for all plots where ($\bar{x}$, $\bar{y}$) departed significantly from (0, 0); and 3) comparing the two mean angles to see if the aspect-to-sprout vector angle was smaller than expected for randomly chosen vectors. A small aspect-to-sprout vector would suggest that sprouts are centered on the downslope side of the parent. To test for an interaction between percent slope and sprout spatial distribution, slope was correlated with the Euclidean distance between ($\bar{x}$, $\bar{y}$) and (0, 0) using only those cases where sprout centroids were not centered on the parent and also using all 31 plots.

Comparison of superficial root density with sprout density was not possible for individual plots because of high variability in root density samples. Root length and root surface area estimates, when analyzed for appropriate sample size using Stein's two stage sample technique (Steel and Torrie, 1960), revealed that a complete enumeration of superficial woody root length or surface area was needed to obtain estimates within 10% of the mean. Complete enumerations were impractical and thus data from 1,092 quadrats (21 plots) were combined. This composite sample was then compared to

the combined density of sprout clumps in the same 21 plots.

To assess influence of parent vigor on patterns of sprouting, the 21 trees sampled in 1983 and 1984 were divided into low- and high-vigor groups. The two groups were then compared to test for differences in sprout parameter estimates such as mean distance of sprouts from the parent and mean depth of sprout origin. Parent trees sampled in 1985 were excluded because fewer sprout variables were measured in these plots. Parent vigor groups were determined by: 1) ranking all trees according to crown health index, and then according to 3-yr radial growth (with the healthiest tree receiving a rank of one in both cases); 2) summing the two rankings; and 3) designating the seven trees with the lowest summed ranks as high-vigor trees, the seven trees with the highest summed ranks as low-vigor trees, and the remaining seven trees as intermediate. Parent trees were also divided into 1983 and 1984 groups to determine if sprouting patterns differed by yr of measurement. In each parent group, sprouts were divided into 12 age classes including 1, 2, 3, 4, 5, 6, 7, 8 + 9, 10 + 11, 12 + 13, 14 + 15 + 16, and 17+ years old. For each age class in each group, interval estimates ($\alpha = 0.05$) were constructed for depth of sprout origin, number of stems per clump, ht of the TSPC, extension growth of the TSPC, mean parent root diam, and mean distance of sprouts from the parent. For each variable-sprout age class combination (60 total), interval estimates for the high- and low-vigor groups were inspected for overlap. By chance alone, about 95% of the intervals should overlap. If fewer than 95% overlap, the measured sprout variables may be related to parent vigor (or yr of measurement).

To determine the histological origin of sprouts, roots with young sprouts were collected from several stands and brought to the lab for sectioning. Cross sections of fresh root materials were stained with 1% safranin and studied microscopically.

RESULTS—Excluding the 1-yr old seedlings sampled in 1985, 673 seedlings, 1,061 sprout clumps, and 39 beech of unknown origin were mapped in the 31 plots. The distribution of sprouts and seedlings in a representative sampled plot is shown in Fig. 1. In every plot, sprouts were concentrated near the parent tree. For all 31 plots combined, 95% and 99% of the sprouts were no more than 7.95 and 10.15 m from the parent, respectively. No sprouts originating from any of the sampled parents were found outside of the 24 × 24 m plots. In

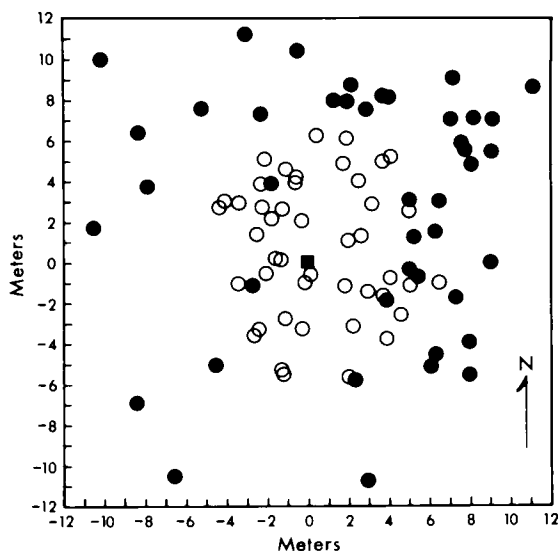


Fig. 1. Distribution of sprouts (open circles) and seedlings (closed circles) around a parent tree (square) in a representative sampled plot.

contrast to sprouts, seedlings were well scattered throughout the plots (Fig. 1). Excluding 1-year old seedlings sampled in 1985, the median seedling to parent distance was 8.41 m. Seedling densities within and outside of the area having 95% of all sprouts (radius = 7.95 m) averaged $0.041/\text{m}^2$ and $0.039/\text{m}^2$, respectively.

Sprout distributions were circular with only six plots having significantly correlated X and Y axes (Table 1). Two-thirds of the correlation coefficients were less than 0.3 while 80% were less than 0.4 and 88% were less than 0.5. Although sprout centroids were significantly offset from the plot center in 12 out of 30 cases, most were offset very slightly. Altogether, 16 were offset by less than 1 m, 10 by 1–2 m, and only 4 by more than 2 m (Table 1).

Mean distance of sprouts from parents did not differ significantly among the 12 sprout age classes ($F = 1.68$, $P = 0.072$); however, 1-yr old sprouts had the smallest mean distance (Table 2). When all sprout age classes were combined, mean sprout to parent distance was significantly correlated with parent DBH ($r = 0.784$, $P < 0.0001$). In general, the number of sprouts was higher around larger trees, but high variability was noted (e.g., one 50.7 cm tree had only 3 sprouts while a 30.2 cm tree had 87 sprouts [Table 1]). Also, larger trees had older sprouts.

A mean angle of 69.62° (range = 1.0° – 127.9°) was formed by the sprout vector ($\bar{x}$, $\bar{y}$) and the aspect of the plot in the 12 trees whose centroids were significantly offset from (0, 0). This

TABLE 1. Spatial distribution patterns of sprouts in 31 plots

Parent DBH (cm)	Number of sprouts	Mean sprout to parent distance (m)	Centroid to parent distance (m) ^a	Correlation of X and Y axes ^a
15.8	8	1.78	0.29	-0.116
16.3	7	2.36	1.48	-0.287
18.8	9	2.23	1.55	0.659
20.6	6	1.82	0.94	0.062
24.6	12	2.03	1.50***	0.168
28.7	30	2.58	0.84	0.142
29.2	28	2.41	0.26	-0.203
29.7	22	3.21	0.83	0.383
30.2	87	2.42	0.79***	0.119
30.7	17	3.65	2.70***	0.552*
30.7	19	3.30	0.32	0.299
31.5	16	3.17	1.37*	0.404
31.5	19	2.03	0.35	0.379
31.7	13	1.95	1.14	0.079
33.2	5	3.11	2.92*	-0.712
35.3	35	3.44	1.34**	0.056
36.2	47	3.13	0.52	0.343*
36.8	50	3.51	1.18***	0.331*
38.4	17	2.46	0.63	-0.003
39.4	13	2.33	0.53	0.232
40.9	44	4.12	1.41*	0.219
41.1	54	4.51	3.60***	0.495***
43.2	41	3.82	0.54	0.077
45.0	53	4.19	0.90*	0.077
46.2	37	4.72	1.18	-0.038
47.8	76	5.12	3.36***	0.391***
50.0	110	3.95	0.75	-0.321***
50.7	3	3.12	N	N
51.3	45	4.11	0.89	-0.078
52.6	60	5.29	1.68***	-0.041
54.6	78	4.07	0.65	0.201

^a Significant values are noted at the 5% (*), 1% (**), and 0.5% (***) probability levels. N = not enough sprouts for test.

angle was greater than the mean angle of 59.87° between 1,000 randomly chosen pairs of vectors. The mean angle for all trees with more than four sprouts (29) was 74.1° . Because the mean aspect-to-sprout vector angle was not less than expected for unrelated vectors, the direction from the parent to the sprout centroid appears to be unrelated to aspect. Since most centroids were not significantly offset from the parent, and since direction of offset and aspect were apparently unrelated, no significant relationship between slope and degree of offset was expected. The latter was confirmed by a correlation coefficient of only -0.112 between plot slope (range = 0–50%) and distance of the sprout centroid from the plot center (range = 0.26–3.60 m).

Using the 21-tree composite sample, sprout density (no./m²), density of root length (cm/m²), and density of root surface area (cm²/m²) at various distances from the parent were scaled

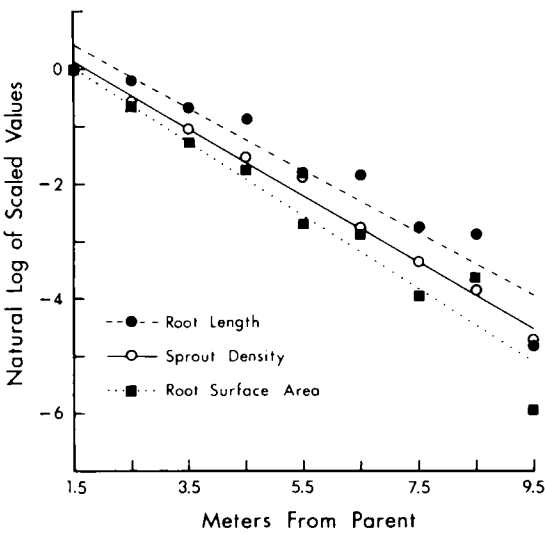


Fig. 2. Sprout density (no./m²), superficial root length (cm/m²), and superficial root surface area (cm²/m²) for various distances from the parent. Data are means from 21 plots, scaled to maximum = 1.000, and then transformed to natural log values. Lines are linear regressions.

(maximum = 1.000) and transformed by taking the natural log of the scaled values. Figure 2 shows an exponential decline for all three variables, as denoted by the linear decline in the log values. Linear regressions of the log values resulted in an estimated slope of -0.578 for sprout density ($r^2 = 0.99$), -0.546 for root length ($r^2 = 0.91$), and -0.654 for root surface area ($r^2 = 0.93$). The similarity among the slopes suggests a close relationship among the three variables.

Parent roots adjacent to the 764 sprouts were generally larger than roots sampled in the 1,092

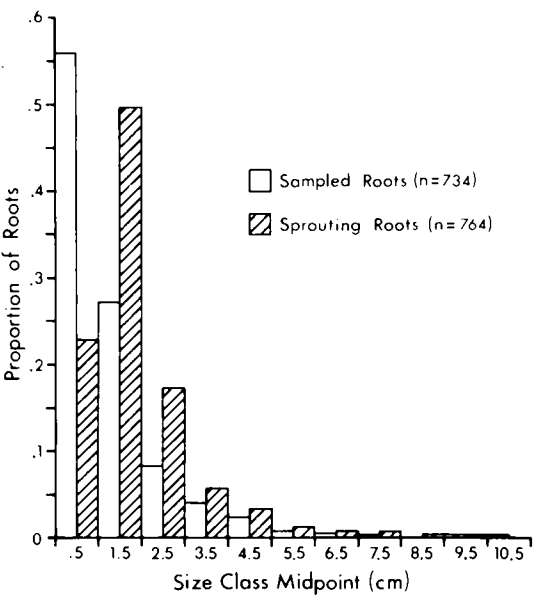


Fig. 3. Diam distributions for roots attached to 764 sprouts and for systematically sampled roots in 21 plots.

quadrats (Fig. 3). Parent root and sampled root diameter distributions were significantly different ($P < 0.01$) according to an approximate Kolmogorov-Smirnov test (Sokal and Rohlf, 1981).

Of the 764 sprouts sampled in 1983–84, 98% arose from roots within the surface organic horizons. The mean depth of origin was 2.91 cm above the surface of the mineral soil, and the mean parent root diam was 1.80 cm. Although significant differences in mean parent root diam and depth of sprout origin were found among sprout age classes, most of the differ-

TABLE 2. Measurements for 764 root sprout loci in 12 age classes

Age class (yrs)	Total ht (cm)	Number of live stems per clump	Parent root diam (cm)	Origin (cm above mineral soil)	Distance from parent (m)
1	9.12 ± 4.03 ^a *	3.10 ± 2.39 ^a	2.18 ± 1.45 ^{ab}	3.64 ± 2.58 ^a	2.99 ± 1.88 ^a
2	17.04 ± 10.21 ^b	3.31 ± 2.66 ^a	1.82 ± 1.35 ^{bc}	3.34 ± 2.43 ^{ab}	3.64 ± 1.81 ^a
3	24.18 ± 11.80 ^a	2.51 ± 2.48 ^{ab}	1.55 ± 1.05 ^{bc}	2.56 ± 2.40 ^{ab}	3.60 ± 2.15 ^a
4	36.76 ± 17.32 ^f	1.80 ± 1.55 ^{bc}	1.33 ± 0.96 ^c	2.06 ± 1.72 ^b	4.04 ± 1.83 ^a
5	47.28 ± 21.25 ^{ef}	1.74 ± 1.08 ^{bc}	1.64 ± 1.34 ^{bc}	3.19 ± 2.51 ^{ab}	3.99 ± 2.59 ^a
6	62.04 ± 23.65 ^{de}	1.71 ± 1.67 ^{bc}	1.70 ± 1.17 ^{bc}	2.87 ± 1.86 ^{ab}	3.76 ± 1.82 ^a
7	69.53 ± 28.92 ^{cd}	1.33 ± 0.79 ^c	1.83 ± 1.56 ^{bc}	2.75 ± 2.18 ^{ab}	3.62 ± 2.12 ^a
8–9	70.82 ± 31.18 ^{cd}	1.26 ± 0.59 ^c	1.56 ± 1.04 ^{bc}	2.57 ± 2.26 ^{ab}	3.79 ± 2.31 ^a
10–11	77.75 ± 31.18 ^{cd}	1.35 ± 0.90 ^c	1.57 ± 0.87 ^{bc}	3.05 ± 2.44 ^{ab}	3.64 ± 2.35 ^a
12–13	91.76 ± 39.25 ^{bc}	1.21 ± 0.33 ^c	1.64 ± 0.80 ^{bc}	2.72 ± 2.08 ^{ab}	3.88 ± 2.50 ^a
14–16	123.78 ± 52.37 ^b	1.32 ± 1.29 ^c	1.75 ± 0.74 ^{bc}	2.47 ± 2.19 ^{ab}	4.18 ± 2.52 ^a
17+	218.00 ± 127.67 ^a	1.14 ± 0.77 ^c	2.73 ± 1.30 ^a	3.34 ± 2.51 ^{ab}	3.45 ± 2.32 ^a
Overall mean	68.74 ± 71.76	1.85 ± 1.77	1.80 ± 1.24	2.91 ± 2.31	3.67 ± 2.17

Numbers are means plus or minus one S.D. Within columns, values followed by the same letter are not significantly different (Tukey's test, $P < 0.05$). Natural log of total ht and sq. root of number of stems per clump were used before analysis to remove heteroscedastic variance.

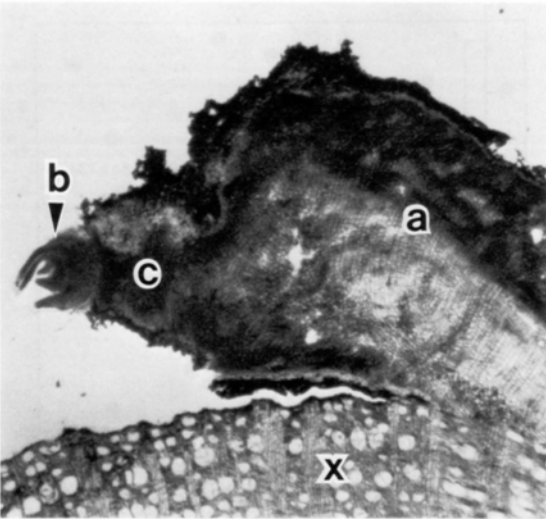


Fig. 4. Cross-section of injured beech root showing callus (c), adventitious bud (b), old xylem (x), and newly initiated cambium (x). $\times 28$.

ences could be attributed to 1-yr old sprouts which frequently develop on top of large exposed roots near the parent (Table 2) (Tukey's test). The TSPC was most often attached to the top (68.5%) or side (28.8%) of the parent root. The proportion of TSPC's attached to the top, side, and bottom was unrelated to sprout age (Table 3) as determined by the G-test ($P = 0.8985$) (Sokal and Rohlf, 1981). Mean number of stems per clump decreased from approx. three to one and total ht increased from 9.12 to 218.00 cm between the TSPC ages of 1 and 17+ (Table 2).

Only 4.3% of the 764 sprouts measured in 1983–84 were disconnected from the parent

stem, and 66.7% of the disconnected sprouts were greater than 10 yrs old (Table 3). However, entire root systems were not excavated to determine if seemingly disconnected sprouts were linked to the parent through root grafts. Root grafting of superficial roots was commonly noted.

When sprouts around the seven least vigorous trees were grouped together and compared to sprouts around the seven most vigorous trees, interval estimates of sprout variables were overlapping in 90.0% of the 60 between-group comparisons. The few cases of non-overlap were scattered and not concentrated in any one sprout age class or variable; however, one trend was noted. The number of sprouts per clump was consistently lower around low-vigor parents from sprout ages one to four. When the 21 trees were divided into 1983 and 1984 groups, 95% of all interval estimates overlapped.

In all examined cases, sprouts originated as adventitious buds that developed from callus tissue around wounded root cambium (Fig. 4). Unexpanded buds up to 2 yrs old were found; however, most buds appeared to die or lose all apical meristem before reaching 3 yrs old (Fig. 5–6). No dormant buds beneath the cork cambium were found.

DISCUSSION—Sprouts were distributed within circles surrounding parent trees. Mean distance of sprouts from parents was greatest for large trees. In all clones, however, most sprouts were less than 8 m from the parent. On average, sprout density was highest between 1 and 2 m from the parent and declined exponentially beyond 2 m. In the few cases where the circle was offset from the parent, the direction and

TABLE 3. Position of sprouts on parent roots and number of sprouts disconnected from the parent for 12 age classes of sprouts

Age class (yrs)	Number of sprouts	Percent of sprouts attached to parent root on				Disconnected from parent	
		Top	Side	Bottom	Unknown	Number	%
1	72	68.1	29.2	2.78	0.0	7	9.3
2	68	58.8	39.7	1.47	0.0	2	2.9
3	77	66.2	31.2	2.60	0.0	0	0.0
4	44	61.4	31.8	6.82	0.0	0	0.0
5	50	60.0	38.0	2.00	0.0	0	0.0
6	85	67.1	31.8	1.18	0.0	0	0.0
7	76	56.6	42.1	1.32	0.0	1	1.3
8–9	78	55.1	44.9	0.00	0.0	1	1.3
10–11	63	65.1	33.3	0.00	1.6	3	4.8
12–13	41	53.7	43.9	2.44	0.0	2	4.9
14–16	41	56.1	43.9	0.00	0.0	3	7.3
17+	69	75.4	23.2	1.45	0.0	14	20.3
Total	764	—	—	—	—	33	—
Overall mean	—	62.6	35.6	1.7	0.1	—	4.3

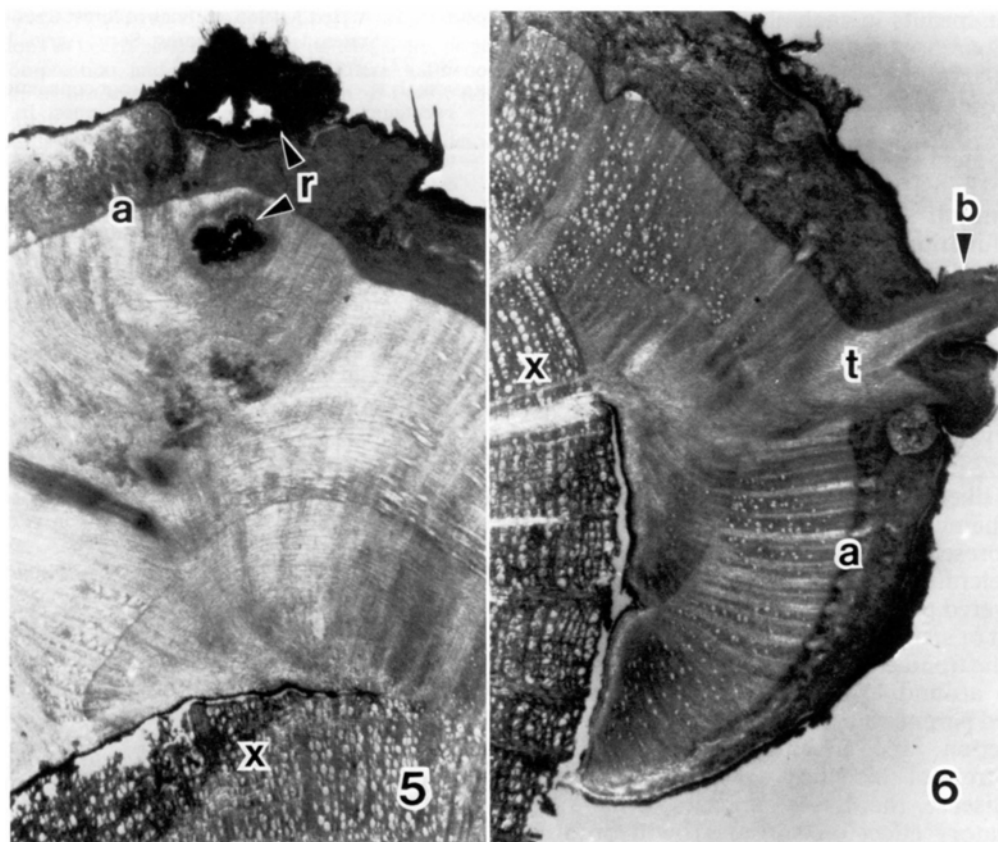


Fig. 5, 6. Cross-sections of beech roots 2 yrs after injury showing cambium initiated after injury (a), xylem formed before injury (x), remains of dead bud overgrown in second growing season after injury (r), vascular bud trace (t), and living adventitious bud (b). $\times 15$.

degree of offset was apparently unrelated to slope or aspect. Sprout age classes were well-dispersed throughout each clone, and thus no wave-like movement or expansion of the clone is indicated.

The small total area occupied by sprouts, the high concentration of sprouts relatively close to the parent stem, and the apparent lack of wave-like expansion of clones suggest that root sprouting may contribute more to the persistence of beech at a given location than to lateral spread to adjacent sites, except over many generations of trees. If this is true, seed dispersal rather than clonal spread may account for most colonization of stands by beech, and stands so colonized would be composed of many small clones rather than fewer, larger clones. A similar pattern of restricted clonal spread with more effective spread by seed exists for *Alnus rugosa* (Du Roi) Spreng. (Huenneke, 1985).

The presence of root sprouts is apparently a function of superficial root density, as suggested by Fig. 2, and also of root injury as suggested by Ward (1961). If root density and

frequency of injury are important, variation in these two factors may explain: 1) large differences in the number of sprouts among clones with similar parent tree DBH; and 2) regional and within-region variability in the relative abundance of sprouts observed by other workers (Ward, 1961; Bormann et al., 1970; Held, 1983). This contention is supported by one study where sprouting was observed in shallowly but not deeply rooted clones (Logan, 1959). The causes and densities of root injury were not directly measured in the present study; however, root injury was noted on most of the woody superficial roots inspected. Some injuries were found where roots were scraped by rocks or eaten by animals (the latter usually occurred around animal burrows), but most sprouts originated on the tops of roots where little or no contact with rocks was apparent. Also, frequent sprouting was noted in some clones growing in soils with relatively few coarse fragments. The suggestion that freezing and thawing may injure roots and lead to sprouting (Held, 1983) merits further investigation.

Most sprouts in each clone were connected to the parent stem. In two clones with high-vigor parent trees, development of adventitious roots at the base of the sprout and senescence of the parent root proximal to the parent stem were common among sprouts greater than 10 yrs old. Studies are needed to determine if this tendency for sprouts to become functionally independent of the parent stem is: 1) genetically, physiologically, or environmentally controlled; and 2) related to the success of clones to persist at a site for many generations.

That roots attached to sprouts were larger in diam than a systematic sample of all woody roots (Fig. 3) suggests that: 1) some threshold root size must be reached before sprouting becomes likely; 2) sprouts formed on small roots are ephemeral with very low survivorship; or 3) the presence of sprouts on a root is correlated to accelerated root diam growth.

Lowered parent vigor was uncorrelated with total ht or spatial distribution of sprouts. However, the trend of fewer sprout stems per sprout clump around low-vigor trees suggests that lowered parent vigor may reduce annual sprout production. Since much of the lowered vigor in the trees of this study was caused by beech bark disease, the disease appears to have no stimulatory effect on sprout growth or production (and possibly a negative effect on production) as long as the parent tree is alive.

An important finding of this study was the close association of new sprout initiation with injury, callus tissue formation, and adventitious bud development on superficial roots. The relatively brief life span of adventitious buds suggests that frequent root injuries are needed to assure annual production of new sprouts. Potential sources of injury include mammals, freeze-thaw action, and human activity. Control of such sources may help to regulate beech regeneration by sprouting.

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